

ISTQB® CTFL v4.0

Chapter 3

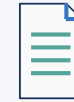
Static Testing

Main reference: ISTQB CTFL syllabus. Speaker notes expanded with Stapp textbook explanations.



Chapter 3 learning route

Syllabus numbering retained: 3.1 → 3.2



3.1 Static Testing Basics

FL-3.1.1 (K1) Recognize types of products that can be examined by the different static test techniques

FL-3.1.2 (K2) Explain the value of static testing

FL-3.1.3 (K2) Compare and contrast static and dynamic testing

3.2 Feedback and Review Process

FL-3.2.1 (K1) Identify the benefits of early and frequent stakeholder feedback

FL-3.2.2 (K2) Summarize the activities of the review process

FL-3.2.3 (K1) Recall which responsibilities are assigned to the principal roles when performing reviews

FL-3.2.4 (K2) Compare and contrast the different review types

FL-3.2.5 (K1) Recall the factors that contribute to a successful review



INTERACTIVE: Ask learners to name one document they review before testing software.

3.1 Static Testing Basics

Static testing evaluates work products without executing the software.



Manual examination

- Reviews of requirements, user stories, designs, code, tests and documents
- Stakeholders discuss and improve work products



Tool-supported analysis

- Static analysis of code, models or structured text
- Can be automated in CI pipelines



Objectives

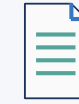
- Detect defects early
- Assess readability, completeness, correctness, testability and consistency
- Support verification and validation



INTERACTIVE: Ask: can a user story refinement meeting be a static testing activity?

3.1.1 Work products examinable by static testing

Almost any readable work product can be reviewed.



Common review targets

- requirements
- source code
- test plans and test cases
- product backlog items



Other examples

- test charters
- project documentation
- contracts
- models



Limits

- Static analysis needs a formal structure
- Some third-party executables may be inappropriate for review or analysis



INTERACTIVE: Ask learners to sort work products into “review only” vs “tool-analyzable”.

3.1.2 Value of static testing

Static testing finds defects early and supports shared understanding.



Early defect detection

- Finds defects before execution is possible
- Prevents defects from propagating to later work products



Defects dynamic testing may miss

- unreachable code
- ambiguous requirements
- design patterns not implemented as intended
- non-executable work product defects



Collaboration value

- Builds shared understanding
- Improves communication
- Supports confidence in work products



INTERACTIVE: Ask: why can reviews lower total project cost even if they take time?

3.1.3 Static testing vs dynamic testing

They complement each other and find different kinds of problems.



Static testing

- Software does not need to run
- Reviews documents/code
- Can begin before software exists
- Finds defects directly

Example: review login requirements



Dynamic testing

- Software must be executed
- Executes test cases
- Usually needs executable software
- Often detects failures caused by defects

Example: actually try to log in



INTERACTIVE: Ask learners to give one defect easier to find statically and one easier dynamically.



3.2.1 Benefits of early and frequent stakeholder feedback

Feedback prevents “building the wrong thing for too long.”



Earlier understanding

- Communicate quality problems early
- Prevent requirement misunderstandings
- Keep changing requirements visible

Less rework

- Reduce costly late corrections
- Lower risk of missed deadlines
- Reduce blame games and project failure risk

Better focus

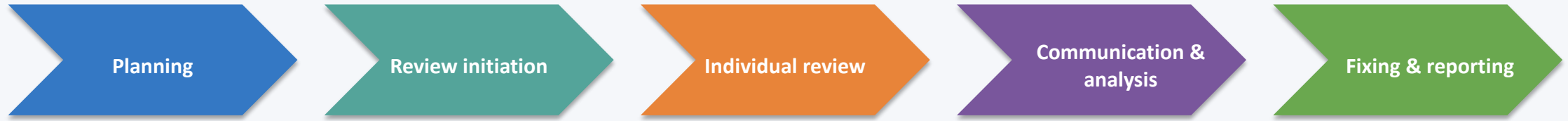
- Improve understanding of what is being built
- Prioritize high-value features
- Focus on features with positive risk impact



INTERACTIVE: Ask learners to recall a case where late stakeholder feedback caused rework.

3.2.2 Review process activities

The generic review process is structured but flexible.



Large work products may require the review process to be invoked several times. More formal reviews use more of the defined tasks.



INTERACTIVE: Ask: why is individual preparation important before a review meeting?

3.2.2 Review process activities: Planning & Review Initiation



First define the review; then make sure people and materials are ready.

Planning — define the review

- Purpose and work product
- Quality characteristics and focus areas
- Exit criteria and supporting information
- Effort and timeframe

Review initiation — prepare to start

- Everyone can access the correct work product
- Participants understand roles and responsibilities
- Required standards, checklists and supporting materials are available



INTERACTIVE: Which activity covers exit criteria? Which activity checks that everyone has the correct document?

3.2.2 Review process activities: Individual Review → Analysis



→ Fixing

Find possible problems, decide what they mean, then correct and report.

Individual review

- Each reviewer assesses quality independently
- Use techniques such as checklist- or scenario-based review
- Log anomalies, recommendations and questions

Communication & analysis

- An anomaly is not automatically a defect
- Decide status, ownership and required action
- Assess overall quality and follow-up needs

Fixing & reporting

- Create defect reports for confirmed defects
- Track corrective actions
- Accept when exit criteria are reached
- Report review results



INTERACTIVE: Why should an anomaly be discussed before it is classified as a defect?



3.2.3 Roles and responsibilities in reviews

Reviews involve multiple roles; one person may hold more than one role.



Manager / Review leader

- Manager provides resources and decides what is reviewed
- Review leader organizes people, time and place



Author / Reviewer

- Author creates and fixes the work product
- Reviewers perform the review and identify anomalies



Moderator / Scribe

- Moderator facilitates meetings and safe discussion
- Scribe records anomalies, decisions and review information



INTERACTIVE: Ask learners which role should not dominate anomaly discussion and why.

3.2.4 Review types

Review formality increases from informal review to inspection.



Informal review

no defined process; anomaly detection

Walkthrough

author-led; learning and consensus

Technical review

technically qualified reviewers; moderator-led

Inspection

most formal; maximum anomaly detection

Formality may depend on SDLC, process maturity, criticality, complexity, legal/regulatory needs and audit-trail requirements.



INTERACTIVE: Choose the most suitable review type for a safety-critical design document.

3.2.4 Review types: Informal Review vs Walkthrough

A quick check differs from an author-led explanation and discussion.



Informal review — QUICK CHECK

- No defined process
- No formal documented output required
- Main objective: detect anomalies
- Example: ask a colleague to quickly check test cases before use

Walkthrough — AUTHOR LEADS

- The author presents and explains the work product
- Individual review beforehand is optional
- Can educate, build consensus, generate ideas and detect anomalies



INTERACTIVE: If the author explains the requirements step by step to the team, which review type is it?

3.2.4 Review types: Technical Review vs Inspection

Technical experts decide; inspection applies the most formal review discipline.



Technical review — EXPERTS DECIDE

- Performed by technically qualified reviewers
- Led by a moderator
- Gain consensus and make decisions on technical problems
- Also detects anomalies and builds confidence

Inspection — MOST FORMAL

- Follows the complete generic review process
- Main objective: find the maximum number of anomalies
- Collects metrics for SDLC and inspection improvement
- Author cannot be review leader or scribe



INTERACTIVE: Which review type explicitly collects metrics and restricts the author's roles?

3.2.5 Success factors for reviews

Effective reviews require preparation, objectives, culture and follow-up.



Clear purpose

- Define objectives and measurable exit criteria
- Never evaluate participants as a review objective



Right scope and type

- Choose suitable review type
- Review small chunks
- Give adequate preparation time



Supportive culture

- Management support
- Training for roles
- Safe facilitation and feedback for improvement



INTERACTIVE: Ask learners to identify one factor missing in failed review meetings they have seen.

Chapter 3 recap and checkpoint

Check the review and static testing essentials.

3



Checkpoint questions

- What makes static testing different from dynamic testing?
- Which work products can be reviewed?
- Why does early stakeholder feedback matter?
- What are the five generic review process activities?
- How do informal review, walkthrough, technical review and inspection differ?



INTERACTIVE: Ask learners to create one review checklist item for a user story.